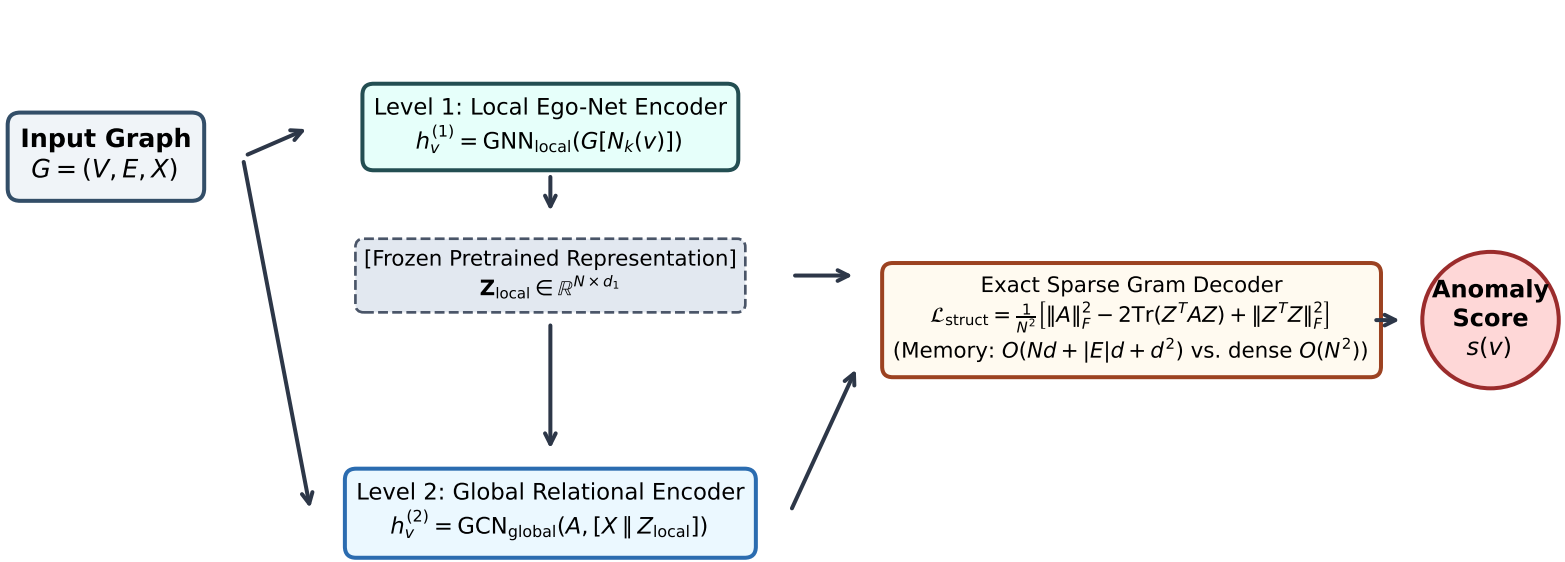
Decoupled Local-to-Global (DLG) Architecture & Exact Sparse Gram Decoder
Input Graph
$G = (V, E, X)$
Level 1: Local Ego-Net Encoder
$h_v^{(1)} = \mathrm{GNN}_{\mathrm{local}}(G[N_k(v)])$
[Frozen Pretrained Representation]
$\mathbf{Z}_{\mathrm{local}} \in \mathbb{R}^{N \times d_1}$
Level 2: Global Relational Encoder
$h_v^{(2)} = \mathrm{GCN}_{\mathrm{global}}(A, [X \,\|\, Z_{\mathrm{local}}])$
Exact Sparse Gram Decoder
$\mathcal{L}_{\mathrm{struct}} = \frac{1}{N^2}\left[\|A\|_F^2 - 2\mathrm{Tr}(Z^T A Z) + \|Z^T Z\|_F^2\right]$
(Memory: $O(Nd + |E|d + d^2)$ vs. dense $O(N^2)$)
Anomaly Score
$s(v)$